Mathematical and Computational Methods

Unit 2: Complex numbers I: algebra and geometry

In Unit 1 we built the chain of number systems and noticed that it could not stop at the reals: the simple equation $x^2 + 1 = 0$ has no real solution, because the square of a real number is never negative. This unit takes the final step, $\mathbb{R} \subset \mathbb{C}$. We introduce the imaginary unit, do arithmetic with complex numbers, meet the conjugate and the modulus, picture everything on the *Argand diagram*, and finish with the *polar form*, in which multiplication becomes rotation and scaling.

Where this fits. This completes the number systems of Unit 1 and leans directly on its trigonometry: radians, the values of sin and cos, and the addition formulas all return when we develop the polar form. The geometric view of multiplication established here is the springboard for Unit 3, where the exponential (Euler) form, De Moivre's theorem and complex roots make these ideas effortless.

Contents

Learning objectives

By the end of this unit you will be able to:

- Do complex arithmetic in Cartesian form using the conjugate and modulus.
- Solve real quadratics with complex (conjugate-pair) roots.
- Represent complex numbers on the Argand diagram and add them as vectors.
- Convert between Cartesian and polar form; multiply and divide in polar form as rotation and scaling.

1 The algebra of complex numbers

1.1 The imaginary unit

To solve $x^2 + 1 = 0$ we need a number whose square is -1. No real number qualifies, so we introduce a new one.

Definition 1.1 (Imaginary unit and complex numbers). The *imaginary unit* i is a symbol obeying the single rule

$$\mathrm{i}^2 = -1.$$

A *complex number* is an expression $z = a + b\,\mathrm{i}$ with $a, b \in \mathbb{R}$. The real number $a = \mathrm{Re}(z)$ is the *real part* and $b = \mathrm{Im}(z)$ is the *imaginary part* (itself a real number). The set of all complex numbers is denoted $\mathbb{C}$. Writing z as $a + b\,\mathrm{i}$ is called its *Cartesian form*.

A real number is the special case $b = 0$, so $\mathbb{R} \subset \mathbb{C}$; a number with $a = 0$ and $b \neq 0$, such as 3i, is called *purely imaginary*. Two complex numbers are equal exactly when they match in both parts.

Definition 1.2 (Equality). $a + b\,\mathrm{i} = c + d\,\mathrm{i}$ if and only if $a = c$ *and* $b = d$. A single complex equation thus encodes *two* real equations.

Note (Why $\mathrm{i}^2 = -1$ and not $\mathrm{i} = \sqrt{-1}$). It is tempting to write $\mathrm{i} = \sqrt{-1}$, but this invites contradiction: blindly applying $\sqrt{x}\sqrt{y} = \sqrt{xy}$ would give $1 = \sqrt{1} = \sqrt{(-1)(-1)} = \sqrt{-1}\,\sqrt{-1} = \mathrm{i} \cdot \mathrm{i} = -1$. The rule $\sqrt{x}\sqrt{y} = \sqrt{xy}$ holds only for non-negative reals. We therefore *define* i by the property $\mathrm{i}^2 = -1$ and never manipulate $\sqrt{-1}$ directly.

Because $\mathrm{i}^2 = -1$, the higher powers of i cycle with period four. From $\mathrm{i}^2 = -1$ we get $\mathrm{i}^3 = \mathrm{i}^2 \cdot \mathrm{i} = -\mathrm{i}$ and $\mathrm{i}^4 = (\mathrm{i}^2)^2 = 1$, after which the pattern repeats:

$$\mathrm{i}^0 = 1, \quad \mathrm{i}^1 = \mathrm{i}, \quad \mathrm{i}^2 = -1, \quad \mathrm{i}^3 = -\mathrm{i}, \quad \mathrm{i}^4 = 1, \quad \mathrm{i}^5 = \mathrm{i}, \ \ldots$$

Example 1.1 ***(Powers of the imaginary unit).*** To evaluate a power of i, divide the exponent by 4 and keep only the remainder (this is called *reducing it modulo 4*). Since $2025 = 4 \cdot 506 + 1$,

$$\mathrm{i}^{2025} = \mathrm{i}^1 = \mathrm{i}.$$

Negative powers follow from $\mathrm{i}^{-1} = \dfrac{1}{\mathrm{i}} = \dfrac{\mathrm{i}}{\mathrm{i}^2} = -\mathrm{i}$.

1.2 Addition, subtraction and multiplication

We extend the arithmetic of $\mathbb{R}$ to $\mathbb{C}$ by requiring the familiar laws—commutativity, associativity and distributivity—to keep holding, together with the single new rule $\mathrm{i}^2 = -1$. These requirements leave no freedom: they *force* the formulas below, which we therefore take as the definition of the three operations.

Definition 1.3 (The three operations). For $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$,

$$\begin{aligned} z + w &= (a + c) + (b + d)\,\mathrm{i}, \\ z - w &= (a - c) + (b - d)\,\mathrm{i}, \\ z\,w &= (ac - bd) + (ad + bc)\,\mathrm{i}. \end{aligned}$$

Addition and subtraction are part by part. The product is just the bracket expansion, with i^2 replaced by -1:

$$(a + b\mathrm{i})(c + d\mathrm{i}) = ac + ad\,\mathrm{i} + bc\,\mathrm{i} + bd\,\mathrm{i}^2 = (ac - bd) + (ad + bc)\,\mathrm{i}.$$

Example 1.2 *(Cartesian arithmetic)*. Let $z = 2 + 3\mathrm{i}$ and $w = 1 - 4\mathrm{i}$. Then

$$z + w = 3 - \mathrm{i}, \qquad z - w = 1 + 7\mathrm{i},$$

and, using the product rule with $a = 2,\ b = 3,\ c = 1,\ d = -4$,

$$z\,w = (2 \cdot 1 - 3 \cdot (-4)) + (2 \cdot (-4) + 3 \cdot 1)\,\mathrm{i} = (2 + 12) + (-8 + 3)\,\mathrm{i} = 14 - 5\mathrm{i}.$$

Remark. The square of a complex number need not be a non-negative real number—indeed it need not be real at all. For example $(2 + 3\mathrm{i})^2 = 4 + 12\mathrm{i} + 9\mathrm{i}^2 = -5 + 12\mathrm{i}$. This is why the real-number test "a square is ≥ 0" has no analogue in $\mathbb{C}$.

1.3 The complex conjugate

Definition 1.4 (Conjugate). The *complex conjugate* of $z = a + b\,\mathrm{i}$ is $\bar{z} = a - b\,\mathrm{i}$: the sign of the imaginary part is reversed.

Conjugation interacts cleanly with every operation. The following are checked by a one-line calculation each.

Proposition 1.1 (Properties of the conjugate). For all $z, w \in \mathbb{C}$, writing $z = a + b\mathrm{i}$,

$$\bar{\bar{z}} = z, \qquad \overline{z + w} = \bar{z} + \bar{w}, \qquad \overline{z\,w} = \bar{z}\,\bar{w},$$

$$z + \bar{z} = 2\,\mathrm{Re}(z), \qquad z - \bar{z} = 2\ \mathrm{Im}(z)\,\mathrm{i}, \qquad z\,\bar{z} = a^2 + b^2.$$

Moreover z is real $\iff$ $\bar{z} = z$, and $\bar{z} = -z$ exactly when $\mathrm{Re}(z) = 0$ (that is, z purely imaginary or zero).

The last identity is the most useful: the product of a number with its conjugate is always a *non-negative real number*,

$$z\,\overline{z} = (a+b\mathrm{i})(a-b\mathrm{i}) = a^2 - (b\mathrm{i})^2 = a^2 + b^2.$$

This single fact powers both the modulus and division below.

1.4 The modulus

Definition 1.5 (Modulus). The *modulus* (or absolute value) of $z = a + b\,\mathrm{i}$ is the non-negative real number

$$|z| = \sqrt{a^2+b^2}, \qquad \text{so that} \qquad |z|^2 = z\,\overline{z}.$$

Proposition 1.2 (Properties of the modulus). For all $z, w \in \mathbb{C}$,

$$|z| \geq 0 \text{ with } |z| = 0 \iff z = 0, \quad |\overline{z}| = |z|\,, \quad |z\,w| = |z|\;|w|\,,$$

and the *triangle inequality* $|z+w| \leq |z| + |w|$.

Example 1.3 (*Computing a modulus*). For $z = 12 - 5\mathrm{i}$,

$$|z| = \sqrt{12^2 + (-5)^2} = \sqrt{144+25} = \sqrt{169} = 13.$$

1.5 Why the modulus behaves well

Two of the properties just listed—the multiplicativity $|zw| = |z|\,|w|$ and the triangle inequality—are worth proving, not merely asserting. Both run on the single identity $z\,\overline{z} = |z|^2$ flagged above as the most useful one.

Proposition 1.3 (Multiplicativity of the modulus). For all $z, w \in \mathbb{C}$, $|z\,w| = |z|\;|w|$.

Proof. Square the left-hand side and use $|u|^2 = u\,\overline{u}$ together with $\overline{z\,w} = \overline{z}\,\overline{w}$ (from the conjugate properties):

$$|z\,w|^2 = (z\,w)\,\overline{z\,w} = (z\,w)(\overline{z}\,\overline{w}) = (z\,\overline{z})(w\,\overline{w}) = |z|^2\;|w|^2\,.$$

Both $|z\,w|$ and $|z|\;|w|$ are non-negative, so taking square roots gives $|z\,w| = |z|\;|w|$. ∎

For the triangle inequality we need one small fact: for any $\zeta = p + q\,\mathrm{i}$,

$$\operatorname{Re}\zeta = p \leq \sqrt{p^2+q^2} = |\zeta|\,.$$

Proposition 1.4 (Triangle inequality). For all $z, w \in \mathbb{C}$, $|z+w| \leq |z| + |w|$, with equality exactly when one of z, w is a non-negative real multiple of the other.

Proof. Expand $|z+w|^2$ with $|u|^2 = u\,\bar{u}$ and $\overline{z+w} = \bar{z}+\overline{w}$:

$$|z+w|^2 = (z+w)(\bar{z}+\overline{w}) = z\bar{z} + w\overline{w} + (z\overline{w} + \bar{z}\,w) = |z|^2 + |w|^2 + 2\,\mathrm{Re}(z\overline{w}),$$

since $\bar{z}\,w = \overline{z\overline{w}}$ and $\zeta + \bar{\zeta} = 2\,\mathrm{Re}\,\zeta$. Bounding the middle term by the fact above and then by multiplicativity ($|z\overline{w}| = |z|\,|\overline{w}| = |z|\,|w|$),

$$|z+w|^2 \le |z|^2 + 2\,|z|\,|w| + |w|^2 = \big(|z| + |w|\big)^2.$$

Taking non-negative square roots gives $|z+w| \le |z| + |w|$. Equality forces $\mathrm{Re}(z\overline{w}) = |z\overline{w}|$, i.e. $z\overline{w} = t$ for some real $t \ge 0$. If $w = 0$ there is nothing to prove; otherwise multiplying both sides by w gives $z\,|w|^2 = t\,w$, so $z = \frac{t}{|w|^2}\,w$ is a non-negative real multiple of w. (On the Argand diagram of Section 2 this says that the two arrows point the same way.) ∎

Remark. The triangle inequality is the statement that "the straight path is shortest": the side $|z+w|$ of a triangle cannot exceed the sum of the other two sides $|z|$ and $|w|$. The same inequality, proved by the same expand-and-bound idea, holds for the length (norm) of vectors in $\mathbb{R}^n$, which Unit 4 introduces.

1.6 Division

Division is made possible by the fact that $w\,\overline{w} = |w|^2$ is real: to divide by w, multiply top and bottom by $\overline{w}$, which clears the imaginary part from the denominator.

Theorem 1.1 (Division). For $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i} \ne 0$,

$$\frac{z}{w} = \frac{z\,\overline{w}}{w\,\overline{w}} = \frac{z\,\overline{w}}{|w|^2} = \frac{(ac+bd) + (bc-ad)\,\mathrm{i}}{c^2+d^2}.$$

In particular the reciprocal is $\dfrac{1}{z} = \dfrac{\bar{z}}{|z|^2}$.

With division in hand, the modulus of a quotient follows from multiplicativity.

Corollary 1.1 (Modulus of a quotient). If $w \ne 0$ then $\left|\frac{z}{w}\right| = \dfrac{|z|}{|w|}$.

Proof. Apply multiplicativity of the modulus to the identity $\frac{z}{w} \cdot w = z$: it gives $\left|\frac{z}{w}\right|\,|w| = |z|$, and dividing by $|w| > 0$ finishes. ∎

Example 1.4 ***(Dividing in Cartesian form).*** Compute $\dfrac{-3+4\mathrm{i}}{6-8\mathrm{i}}$. Multiply by the conjugate $6+8\mathrm{i}$ of the denominator:

$$\frac{-3+4\mathrm{i}}{6-8\mathrm{i}} = \frac{(-3+4\mathrm{i})(6+8\mathrm{i})}{(6-8\mathrm{i})(6+8\mathrm{i})} = \frac{-18-24\mathrm{i}+24\mathrm{i}+32\,\mathrm{i}^2}{6^2+8^2} = \frac{-50}{100} = -\tfrac{1}{2}.$$

Note (A common error). Division is *not* done part by part: $\dfrac{a+b\mathrm{i}}{c+d\mathrm{i}} \neq \dfrac{a}{c} + \dfrac{b}{d}\mathrm{i}$. Always multiply by the conjugate of the denominator.

1.7 Quadratic equations with complex roots

We can now keep the promise made at the start of the unit.

Example 1.5 ***(The equation that motivated $\mathbb{C}$).*** $x^2+1=0$ means $x^2=-1$. Since $\mathrm{i}^2=-1$ and $(-\mathrm{i})^2=-1$, the two solutions are $x=\pm\mathrm{i}$. The equation that had no real root has exactly two complex ones.

More generally, a real quadratic $ax^2+bx+c=0$ whose discriminant b^2-4ac is negative has no real roots, but completing the square produces a pair of complex ones. We can do this once and for all.

Theorem 1.2 (The quadratic formula, negative-discriminant case). Let $a,b,c\in\mathbb{R}$ with $a\neq 0$, and suppose the discriminant $D=b^2-4ac$ is *negative*. Then $ax^2+bx+c=0$ has exactly the two complex roots

$$x=\frac{-b\pm\mathrm{i}\sqrt{4ac-b^2}}{2a},$$

which form a conjugate pair with real part $-\dfrac{b}{2a}$ and imaginary parts $\pm\dfrac{\sqrt{4ac-b^2}}{2a}$.

Proof. Divide by a and complete the square:

$$ax^2+bx+c=a\left(x+\tfrac{b}{2a}\right)^2+c-\tfrac{b^2}{4a}=a\left(x+\tfrac{b}{2a}\right)^2-\tfrac{D}{4a},$$

so the equation becomes $\left(x+\tfrac{b}{2a}\right)^2=\dfrac{D}{4a^2}$. When $D<0$ the right-hand side is negative, and because $\mathrm{i}^2=-1$ it is a square after all:

$$\frac{D}{4a^2}=-\frac{4ac-b^2}{4a^2}=\left(\frac{\mathrm{i}\sqrt{4ac-b^2}}{2a}\right)^2,$$

where $4ac-b^2>0$, so only an ordinary real square root appears. Hence $x+\tfrac{b}{2a}=\pm\dfrac{\mathrm{i}\sqrt{4ac-b^2}}{2a}$, and rearranging yields the stated formula. ∎

Note (The same formula you already know). This is not a new formula but the familiar $x=\left(-b\pm\sqrt{D}\right)/2a$ written so as to keep the imaginary unit outside the square root. When $D\geq 0$ the roots are real; when $D<0$ the factor i surfaces and the roots are a conjugate pair. The case $D=0$ gives the single repeated real root $-b/2a$.

Example 1.6 ***(A quadratic with complex roots).*** Solve $x^2+2x+10=0$. Completing the square,

$$(x+1)^2+9=0 \iff (x+1)^2=-9 \iff x+1=\pm 3\mathrm{i},$$

using $(3\mathrm{i})^2 = 9\,\mathrm{i}^2 = -9$. Hence $x = -1 \pm 3\mathrm{i}$.

Example 1.7 ***(Applying the formula).*** Solve $2x^2 + 2x + 5 = 0$. Here $a = 2$, $b = 2$, $c = 5$, so $D = b^2 - 4ac = 4 - 40 = -36 < 0$ and $4ac - b^2 = 36$. The formula gives

$$x = \frac{-2 \pm \mathrm{i}\sqrt{36}}{2 \cdot 2} = \frac{-2 \pm 6\,\mathrm{i}}{4} = -\tfrac{1}{2} \pm \tfrac{3}{2}\mathrm{i}.$$

As a check, the conjugate pair multiplies back to a real factor: $\left(x + \frac{1}{2}\right)^2 + \frac{9}{4} = x^2 + x + \frac{5}{2}$, which is $\frac{1}{2}(2x^2 + 2x + 5)$.

Proposition 1.5 (Conjugate-pair theorem). Let $p(x) = c_n x^n + \cdots + c_1 x + c_0$ be a polynomial with *real* coefficients $c_0, \ldots, c_n \in \mathbb{R}$. If $\alpha \in \mathbb{C}$ satisfies $p(\alpha) = 0$, then $p(\overline{\alpha}) = 0$ as well: the non-real roots of a real polynomial occur in conjugate pairs.

Proof. Conjugation respects sums and products and fixes real numbers, so $\overline{c_k} = c_k$ and $\overline{\alpha^k} = (\overline{\alpha})^k$. Hence

$$\overline{p(\alpha)} = \overline{\sum_{k=0}^{n} c_k \alpha^k} = \sum_{k=0}^{n} \overline{c_k}\,\overline{\alpha}^k = \sum_{k=0}^{n} c_k (\overline{\alpha})^k = p(\overline{\alpha}).$$

Since $p(\alpha) = 0$, this gives $p(\overline{\alpha}) = \overline{0} = 0$. ∎

Pairing a non-real root α with $\overline{\alpha}$ explains why such roots leave a *real* quadratic factor behind:

$$(x - \alpha)(x - \overline{\alpha}) = x^2 - (\alpha + \overline{\alpha})x + \alpha\overline{\alpha} = x^2 - 2\,\mathrm{Re}(\alpha)\,x + |\alpha|^2,$$

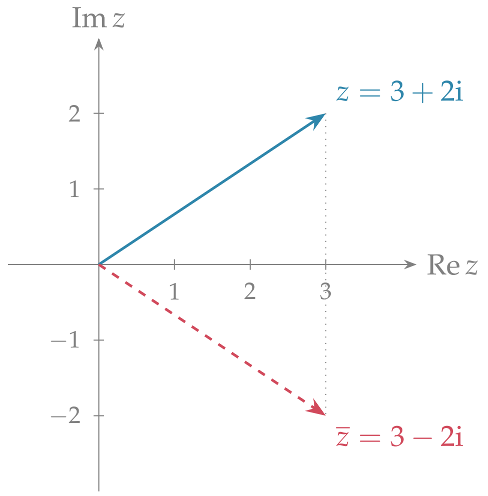
whose coefficients are real. For $\alpha = -1 + 3\mathrm{i}$ this reads $x^2 + 2x + 10$, recovering the quadratic above.

Remark (The fundamental theorem of algebra). Every degree-n polynomial—real or complex coefficients—has exactly n roots in $\mathbb{C}$ when these are *counted with multiplicity* (so a repeated root such as the double root of $x^2 - 2x + 1 = (x - 1)^2$ counts twice). This is the fundamental theorem of algebra, which we return to in Unit 3.

2 The Argand diagram

2.1 Complex numbers as points and vectors

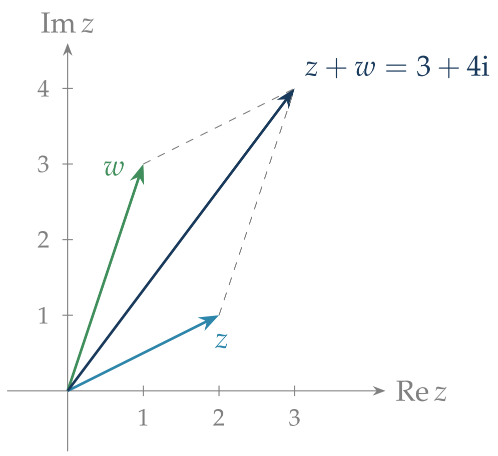
Because a complex number $z = a + b\,\mathrm{i}$ is determined by the real pair (a, b), it can be drawn as a point—or as the arrow from the origin to that point—in the plane. The horizontal axis carries the real part and the vertical axis the imaginary part; the plane used this way is the *Argand diagram* (or complex plane).



The picture immediately makes the modulus and the conjugate geometric: $|z|$ is the *length* of the arrow (by Pythagoras), and $\bar{z}$ is the *reflection of z in the real axis*. Likewise $-z$ is the reflection of z through the origin.

2.2 Addition and subtraction as vectors

Since complex addition adds real and imaginary parts separately, it behaves exactly like vector addition in the plane $\mathbb{R}^2$ (pairs of real numbers, the subject of Unit 4): $z + w$ is found by the tip-to-tail (parallelogram) rule.



Subtraction has an equally useful reading: $z - w$ is the vector *from w to z*, so $|z - w|$ is the *distance between the two points* z and w in the plane. This turns many geometric statements into short algebraic ones.

Example 2.1 ***(Sum and distance).*** For $z = 2 + \mathrm{i}$ and $w = 1 + 3\mathrm{i}$ we have $z + w = 3 + 4\mathrm{i}$, drawn above. The distance between the points is

$$|z - w| = |(2-1) + (1-3)\mathrm{i}| = |1 - 2\mathrm{i}| = \sqrt{1^2 + (-2)^2} = \sqrt{5}.$$

Remark (A data-science reading). Identifying $z = a + b\,\mathrm{i}$ with the point (a, b), the modulus $|z - w|$ is exactly the *Euclidean distance* between two data points in the plane—the same distance that measures how alike two records are in machine learning. The plane-geometry of complex numbers is therefore the two-dimensional case of the $\mathbb{R}^n$ distance developed in Unit 4; multiplication, which we turn to next, adds a rotation that has no real-number counterpart and reappears as the phase of a signal in Unit 3.

2.3 Conjugation, modulus and the absence of order

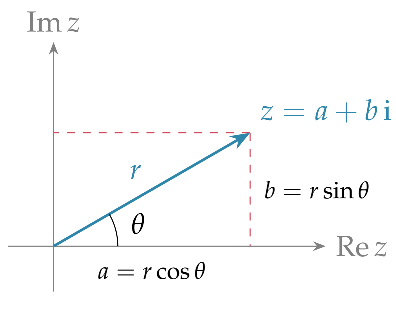
Collecting the geometry: $|z|$ is the distance from the origin, $\bar{z}$ is the mirror image in the real axis (so $|\bar{z}| = |z|$), and addition follows the parallelogram rule. One familiar feature of the real numbers, however, is lost.

Remark (The complex numbers cannot be ordered). On the real line any two numbers satisfy exactly one of $<, =, >$, and the order obeys two rules that make it useful: adding the same number to both sides keeps an inequality, and the product of two positive numbers is positive. No ordering of $\mathbb{C}$ can obey both. If one did, then any $x \neq 0$ would have $x > 0$ or $-x > 0$, and either way the second rule gives $x^2 = (-x)^2 > 0$; applying this to $x = \mathrm{i}$ gives $\mathrm{i}^2 = -1 > 0$, while applying it to $x = 1$ gives $1 = 1^2 > 0$, i.e. $-1 < 0$. These contradict one another. So expressions like "$z < w$" are meaningless for non-real complex numbers—only the real quantities $|z|$, $\mathrm{Re}(z)$, $\mathrm{Im}(z)$ can be compared.

3 The polar form

3.1 Modulus and argument

A point in the plane can be located by Cartesian coordinates (a, b) or, just as well, by how *far* it is from the origin and in which *direction*. For a nonzero z, write $r = |z|$ for the distance and let θ be the angle the arrow makes with the positive real axis, measured anticlockwise in radians (the *argument* of z). Reading off the right-angled triangle—the radians and standard values of Unit 1, now read round a circle—gives



Definition 3.1 (Polar form). Every nonzero $z \in \mathbb{C}$ can be written in *polar form*

$$z = r(\cos\theta + \mathrm{i}\sin\theta), \qquad r = |z| = \sqrt{a^2 + b^2} > 0, \qquad \theta = \arg(z),$$

where $a = r\cos\theta$ and $b = r\sin\theta$. The angle θ is only determined up to whole turns: if θ is one argument of z, then so is $\theta + 2\pi k$ for every $k \in \mathbb{Z}$, and all describe the same number;

$\arg(z)$ denotes any one of them. The unique value in $(-\pi, \pi]$ is the *principal argument* $\operatorname{Arg}(z)$. (For $z = 0$ the modulus is 0 and the argument is undefined.)

The radius is found from $r = \sqrt{a^2 + b^2}$, and the angle from $\tan\theta = b/a$—but the tangent alone does not fix θ, since tan has period π (Unit 1). The Argand diagram decides which quadrant θ lies in. (When $a = 0$ the tangent is undefined: the point is on the imaginary axis, and $\theta = \frac{\pi}{2}$ if $b > 0$, $\theta = -\frac{\pi}{2}$ if $b < 0$.)

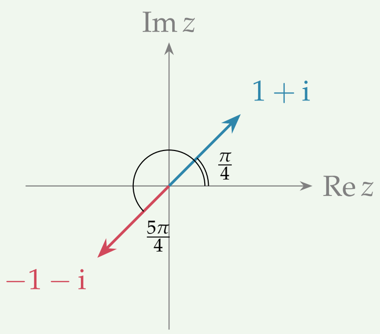
Note (Mind the quadrant). The numbers $1 + \mathrm{i}$ and $-1 - \mathrm{i}$ both give $\tan\theta = 1$, yet they lie in opposite quadrants. A calculator's $\arctan(1) = \frac{\pi}{4}$ is correct only for the first-quadrant point; for $-1 - \mathrm{i}$ the argument is $\frac{\pi}{4} + \pi = \frac{5\pi}{4}$ (equivalently $-\frac{3\pi}{4}$ as a principal value). Always sketch the point first.

3.2 Converting between Cartesian and polar form

Example 3.1 ***(Cartesian → polar, with quadrant care).*** Convert $z = 1 + \mathrm{i}$ and $w = -1 - \mathrm{i}$. Both have $r = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\tan\theta = 1$, but the diagram below separates them: z sits in the first quadrant at $\theta = \frac{\pi}{4}$, while w sits in the third at $\theta = \frac{5\pi}{4}$. Hence

$$z = \sqrt{2}\left(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}\right), \qquad w = \sqrt{2}\left(\cos\tfrac{5\pi}{4} + \mathrm{i}\sin\tfrac{5\pi}{4}\right).$$

Here $\frac{5\pi}{4}$ is *an* argument of w; its principal value is $\operatorname{Arg}(w) = -\frac{3\pi}{4}$.



The two numbers above both make a 45° angle with the real axis; a less symmetric number exercises the full recipe.

Example 3.2 ***(Cartesian → polar in the second quadrant).*** Convert $z = -\sqrt{3} + \mathrm{i}$. The modulus is $r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3 + 1} = 2$. The reference angle—the acute angle the arrow makes with the real axis—comes from the side lengths, $\arctan\dfrac{1}{\sqrt{3}} = \dfrac{\pi}{6}$. The point lies in the *second* quadrant (real part negative, imaginary part positive), and the reference angle is the one made with the negative real axis, so the argument—always measured from the *positive* real axis—is π minus it:

$$\operatorname{Arg}(z) = \pi - \tfrac{\pi}{6} = \tfrac{5\pi}{6}, \qquad z = 2\left(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6}\right).$$

Feeding the raw ratio $b/a = 1/(-\sqrt{3})$ to a calculator returns $\arctan(-1/\sqrt{3}) = -\frac{\pi}{6}$, a *fourth*-quadrant angle—the wrong place. The sketch, not the tangent, fixes the quadrant.

Conversion the other way is just evaluation.

Example 3.3 (*Polar → Cartesian*). For $z = 4\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$, read off the trigonometric values from Unit 1:

$$z = 4\Big(\tfrac{1}{2} + \mathrm{i}\,\tfrac{\sqrt{3}}{2}\Big) = 2 + 2\sqrt{3}\,\mathrm{i}.$$

3.3 Multiplication and division in polar form

The polar form turns multiplication—awkward in Cartesian coordinates—into a simple rule, and reveals its geometric meaning.

Theorem 3.1 (Products and quotients in polar form). If $z = r(\cos\theta + \mathrm{i}\sin\theta)$ and $w = s(\cos\varphi + \mathrm{i}\sin\varphi)$ with $s \neq 0$, then

$$z\,w = rs\big(\cos(\theta+\varphi) + \mathrm{i}\sin(\theta+\varphi)\big), \qquad \frac{z}{w} = \frac{r}{s}\big(\cos(\theta-\varphi) + \mathrm{i}\sin(\theta-\varphi)\big).$$

Proof. Multiply out and group, then use the addition formulas for cos and sin from Unit 1:

$$\begin{aligned} z\,w &= rs\big[(\cos\theta\cos\varphi - \sin\theta\sin\varphi) + \mathrm{i}(\sin\theta\cos\varphi + \cos\theta\sin\varphi)\big] \\ &= rs\big[\cos(\theta+\varphi) + \mathrm{i}\sin(\theta+\varphi)\big]. \end{aligned}$$

For the quotient, first note that $\frac{1}{w} = \frac{1}{s}\big(\cos(-\varphi) + \mathrm{i}\sin(-\varphi)\big)$, because $s(\cos\varphi + \mathrm{i}\sin\varphi)\cdot\frac{1}{s}(\cos\varphi - \mathrm{i}\sin\varphi) = \cos^2\varphi + \sin^2\varphi = 1$; then multiply z by $\frac{1}{w}$ using the product rule just proved. ∎

Note (Multiplication = scaling and rotation). To multiply two complex numbers, *multiply the moduli and add the arguments*; to divide, *divide the moduli and subtract the arguments*. Geometrically, multiplying by w scales every arrow by $|w|$ and rotates it by $\arg(w)$.

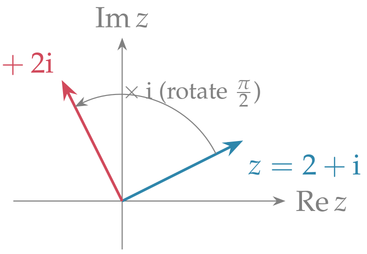
Example 3.4 (*A product in polar form*). Let $z = 2\big(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\big)$ and $w = 3\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$. Then

$$z\,w = 2\cdot 3\Big(\cos\big(\tfrac{\pi}{6}+\tfrac{\pi}{3}\big) + \mathrm{i}\sin\big(\tfrac{\pi}{6}+\tfrac{\pi}{3}\big)\Big) = 6\Big(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2}\Big) = 6\mathrm{i}.$$

Example 3.5 (*A quotient in polar form*). With the same z and w, divide the moduli and subtract the arguments:

$$\frac{z}{w} = \frac{2}{3}\Big(\cos\big(\tfrac{\pi}{6}-\tfrac{\pi}{3}\big) + \mathrm{i}\sin\big(\tfrac{\pi}{6}-\tfrac{\pi}{3}\big)\Big) = \frac{2}{3}\Big(\cos\big(-\tfrac{\pi}{6}\big) + \mathrm{i}\sin\big(-\tfrac{\pi}{6}\big)\Big) = \frac{2}{3}\Big(\tfrac{\sqrt{3}}{2} - \tfrac{1}{2}\mathrm{i}\Big) = \frac{\sqrt{3}-\mathrm{i}}{3}.$$

The cleanest special case is multiplication by i itself. Since $\mathrm{i} = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$ has modulus 1 and argument $\frac{\pi}{2}$, multiplying any z by i leaves its length unchanged and turns it through a quarter turn anticlockwise.



Example 3.6 ***(Multiplication by*** i ***rotates by a quarter turn).*** With $z = 2 + \mathrm{i}$, $\mathrm{i}z = \mathrm{i}(2 + \mathrm{i}) = 2\mathrm{i} + \mathrm{i}^2 = -1 + 2\mathrm{i}$. The modulus is unchanged, $|z| = |\mathrm{i}z| = \sqrt{5}$, and the argument has increased by $\frac{\pi}{2}$, as the figure shows.

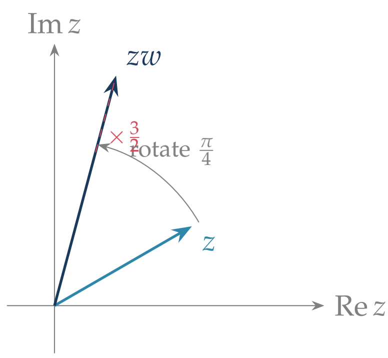
3.4 A general product as rotation and scaling

Multiplying by i is the special case $|w| = 1$, $\arg w = \frac{\pi}{2}$: a pure quarter-turn. The general rule is no harder—multiplying by any w *rotates by* $\arg w$ *and scales by* $|w|$—and the next figure makes that visible for a w that is neither a pure rotation nor a pure scaling.

Example 3.7 ***(A general product, geometrically).*** Take $z = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$ and multiply by w with $|w| = \frac{3}{2}$ and $\arg w = \frac{\pi}{4}$. By the product rule the moduli multiply and the arguments add:

$$|zw| = 2 \cdot \tfrac{3}{2} = 3, \qquad \arg(zw) = \tfrac{\pi}{6} + \tfrac{\pi}{4} = \tfrac{5\pi}{12}.$$

So the arrow for z is rotated anticlockwise through $\arg w = \frac{\pi}{4}$ and stretched by the factor $|w| = \frac{3}{2}$, landing on zw at modulus 3 and argument $\frac{5\pi}{12}$.



The reading reverses for division: dividing by w rotates by $-\arg w$ and scales by $1/|w|$, undoing both moves. In engineering language, multiplying a signal by w applies a phase shift $\arg w$ and a gain $|w|$ at once; the rotation half is exactly the plane rotation that reappears in the geometry of $\mathbb{R}^2$.

Summary

Concept	**Key idea**
Imaginary unit	$\mathrm{i}^2 = -1$; complex number $z = a + b\,\mathrm{i}$ with $\mathrm{Re}(z) = a$, $\mathrm{Im}(z) = b$; powers of i cycle with period 4.
Arithmetic	Add/subtract part by part; multiply by expanding and using $\mathrm{i}^2 = -1$: $zw = (ac - bd) + (ad + bc)\mathrm{i}$.
Conjugate	$\overline{z} = a - b\,\mathrm{i}$; $z\,\overline{z} = a^2 + b^2 = \lvert z\rvert^2$ is real and non-negative.
Modulus	$\lvert z\rvert = \sqrt{a^2 + b^2}$; multiplicative, $\lvert zw\rvert = \lvert z\rvert\,\lvert w\rvert$; triangle inequality.
Division	Multiply top and bottom by $\overline{w}$: $z/w = z\,\overline{w}/\lvert w\rvert^2$.
Argand diagram	$z \leftrightarrow$ point/arrow (a, b); $\lvert z\rvert$ is its length, $\overline{z}$ its reflection in the real axis; addition is the parallelogram rule and $\lvert z - w\rvert$ is a distance.
No order	$\mathbb{C}$ admits no ordering compatible with arithmetic.
Polar form	$z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r = \lvert z\rvert$, $\theta = \arg(z)$ (fix the quadrant from the diagram; principal value $\mathrm{Arg}(z) \in (-\pi, \pi]$).
Polar product	Multiply moduli, add arguments: $zw = rs(\cos(\theta + \varphi) + \mathrm{i}\sin(\theta + \varphi))$; multiplication = scaling + rotation; $\times\mathrm{i}$ is a quarter turn.

Next unit: the exponential (Euler) form $z = r\,\mathrm{e}^{\mathrm{i}\theta}$, De Moivre's theorem, and the powers and roots of complex numbers.

Companion material: a separate *Exercise Sheet 2* (with solutions) develops the practice problems for this unit.